Jiayi (Joanna) Li

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Profile

I develop algebraic, geometric, and statistical foundations for learning systems. My work studies the functions neural networks can represent, the singular geometry and dynamics that select learned solutions, and which claims about those solutions, such as identifiability and robustness, remain meaningful under reparameterization. This program connects algebraic statistics, singular learning theory, and numerical algebraic geometry, with applications to reliable and interpretable AI.

Employment

Max Planck Institute of Molecular Cell Biology and Genetics (MPI-CBG)
Dresden, Germany
Section of Mathematics and Artificial Intelligence 2025–present
Postdoctoral researcher
Mentor: Heather Harrington

Education

University of California, Los Angeles Dec 2025
Ph.D. in Statistics, research focus in Machine Learning Theory
Thesis: *Structure, Symmetry, and Singularity in Learning Systems*
Thesis advisor: Guido Montúfar

The University of Hong Kong & Stony Brook University Jun 2018
B.S. (Hons) Mathematics, Algebraic Geometry
Thesis advisor: Robert Lazarsfeld

Preprints

8. Ben Cullen, Sergio Estan-Ruiz, Riya Danait, **Jiayi Li**. "*A Basin-Selection Perspective on Grokking via Singular Learning Theory*", arXiv:2603.01192, 2026
7. Cecilie Olesen Recke, Sarah Lumpp, Nataliia Kushnerchuk, Janike Oldekop, **Jiayi Li**, Jane Ivy Coons, Elina Robeva. "*Identifiability in Graphical Discrete Lyapunov Models*", arXiv:2601.21818, submitted

Selected Publications

6. Yulia Alexandr, Miles Bakenhus, Maize Curiel, Sameer K. Deshpande, Elizabeth Gross, Yuqi Gu, Max Hill, Joseph Johnson, Bryson Kagy, Vishesh Karwa, **Jiayi Li**, Hanbaek Lyu, Sonja Petrović, Jose Israel Rodriguez. "*New Directions in Algebraic Statistics: Three Challenges from 2023*", Algebraic Statistics, 15, 357–382, 2024
5. Kaie Kubjas, **Jiayi Li**, Maximilian Wiesmann. "*Geometry of Polynomial Neural Networks*", Algebraic Statistics, 15, 295–328, 2024
4. Shuang Liang, Renata Turkeš, **Jiayi Li**, Nina Otter, Guido Montúfar. "*Pull-back Geometry of Persistent Homology Encodings*", Transactions on Machine Learning Research (TMLR), 2024
3. **Jiayi Li**, Yuantong Li, Xiaowu Dai. "*Discussion of 'Estimating Means of Bounded Random Variables by Betting' by Waudby-Smith and Ramdas*", Journal of the Royal Statistical Society: Series B (JRSSB), 86(1), 41–43, 2023
2. Dejun Guo, Xu Jin, Dan Shao, **Jiayi Li**, Yang Shen, Huan Tan. "*Image-Based Regulation of Mobile Robots without Pose Measurements*", IEEE Control Systems Letters (L-CSS), vol. 6, pp. 2156–2161, 2022
1. Ziqi Huang, Yang Shen, **Jiayi Li**, Marcel Fey, Christian Brecher. "*A Survey on AI-Driven Digital Twins in Industry 4.0: Smart Manufacturing and Advanced Robotics*", Sensors, 21(19), 6340, 2021

Leadership & Mentoring

Organizer, Singular Learning Theory Days — Oct 26–27, 2026
MPI-CBG, Dresden; funded by a DFG conference grant

Organizer, AMS Special Sessions at the Joint Mathematics Meetings — 2025–2027
Algebraic Methods in Machine Learning and Optimization
JMM 2025, Seattle; JMM 2026, Washington D.C.
Algebraic, Topological and Geometric Methods in Machine Learning
JMM 2027, Chicago

Minisymposium organizer, SIAM Conference on Mathematics of Data Science (MDS26)
Salt Lake City, UT, USA — Nov 16–20, 2026

Co-mentor with Martin Trapp, LOGML 2026 — Jul 13–17, 2026
Project: *Posterior Geometry and Variational Estimation of Bayesian Complexity*

Mentor, LOGML 2025 — Jul 7–11, 2025
Project: *Symmetry, Degeneracy and Effective Dimensions of Neural Networks*
The project developed into the preprint arXiv:2603.01192

Academic mentor, IPAM Research in Industrial Projects for Students (RIPS)
Project: *Geometry of Representation in Large Language Models* — Jun–Aug 2025
Industrial partner: Microsoft Research

Honors & Grants

ILIAD Fellowship, applied mathematics for AI alignment	2026
Conference grant, German Research Foundation (€4,000)	2026
Dimitris N. Chorafas Foundation Award (USD 10,000)	2024
Dissertation Year Fellowship, UCLA (USD 20,000)	2024
Distinguished Teaching Award Nominee, Statistics/MSOL, UCLA	2024
Summer Mentored Research Fellowship, UCLA (USD 6,000)	2022
ACM Scholarship, Association for Computing Machinery (USD 2,000)	2020
Cathay Bank Scholarship (USD 6,000)	2020
Overseas Research Fellowship, HKU (HKD 12,000)	2016
Undergraduate Research Fellowship, HKU (HKD 15,000)	2015
Travel grants: 20+ awards, over USD 25,000 cumulative	2017–2025

Research Visits

Institute for Computational and Experimental Research in Mathematics (ICERM)
Providence, RI, USA — Jan 19–Apr 23, 2027 (scheduled)
In-residence participant, *Semester Program: Metric Algebraic Geometry — Going Global*

Institute for Pure & Applied Mathematics (IPAM)
Los Angeles, CA, USA — Aug 31–Sep 4, 2026 (scheduled)
Workshop participant, *Foundations of Interpretability*

Max Planck Institute for Mathematics in the Sciences (MPI MiS)
Leipzig, Germany — Summers 2024 and 2025
Visiting Graduate Researcher, Group: Mathematical Machine Learning

Institute for Pure & Applied Mathematics (IPAM)
Los Angeles, CA, USA — Sep–Dec 2024
Graduate Fellow, *Long Program: Mathematics of Intelligences*

Simons Institute for the Theory of Computing
Berkeley, CA, USA — Fall 2024
Short-Term Visitor, *Modern Paradigms in Generalization*

Institute for Mathematical & Statistical Innovation (IMSI)
Chicago, IL, USA — Sep–Dec 2023
Graduate Fellow, *Algebraic Statistics and Our Changing World*

INVITED TALKS

24. Machine Learning in Life Science (MLLS) Seminar
Oct 2, 2026, Copenhagen, Denmark (scheduled)
"Structure, Symmetry and Singularity in Modern Neural Networks"

23. Mathematics Colloquium, Nanjing University
Aug 24, 2026, Nanjing, China (scheduled)
"Structure, Symmetry and Singularity in Modern Neural Networks"

22. Academy of Mathematics and Systems Science, Chinese Academy of Sciences
Aug 12, 2026, Beijing, China
"Structure, Symmetry and Singularity in Modern Neural Networks"

21. Machine Learning Seminar, École Normale Supérieure
May 21, 2026, Paris, France
"Structure, Symmetry and Singularity in Modern Neural Networks"

20. Conference on Geometry, Topology and Machine Learning, Sorbonne Université
May 18–22, 2026, Paris, France
"Basin Selection in Bayesian Learning"

19. Applied CATS Seminar, KTH Royal Institute of Technology
Mar 3, 2026, Stockholm, Sweden
"Grokking as a Phase Transition Between Competing Basins, an SLT Perspective"

18. Mathematical Foundations of AI, Lie–Størmer Colloquium
Dec 3–5, 2025, Simula Research Laboratory, Oslo, Norway
"Structure, Symmetry and Singularity in Learning Systems"

17. Math & Machine Learning Seminar, Boston College
Oct 15, 2025, online
"Theoretical insights for optimizing rational activations"

16. Applied Algebra and Geometry UK Network, Edinburgh meeting
Aug 21, 2025, Edinburgh, UK
"Geometry of Neural Networks with Algebraic Activations"

15. Algebraic Methods in Optimization, ICCOPT
Jul 21–24, 2025, Los Angeles, USA
"Optimizing rational neural networks with trainable parameters"

14. SIAM Conference on Applied Algebraic Geometry
Jul 7–11, 2025, Madison, WI, USA
"Geometry of Neural Networks with Algebraic Activations"

13. Mathematics, AI and Data Science for Material Innovations (MADSMIN), Lancaster University
Jun 9–13, 2025, Lancaster, UK
"Geometry of Neural Networks with Algebraic Activations"

12. Geometry and Machine Learning (GML2025), Sorbonne Université
May 26–30, 2025, Paris, France
"Geometry of Neural Networks with Algebraic Activations"

11. Workshop on the Statistical Theory of Neural Networks
May 5–8, 2025, Egmond aan Zee, Netherlands
"Geometry of Neural Networks with Algebraic Activations"

10. Numerical (Nonlinear) Algebra in the Real World
Feb 6, 2025, Dresden, Germany
"Taming Non-convexity in Shallow Neural Networks with Algebraic Activations"

9. Joint Mathematics Meetings (JMM) 2025
AMS–ASA–SIAM Special Session on Mathematics of Deep Learning: A High-Dimensional Probability Perspective
Jan 11, 2025, Seattle, WA, USA
"Algebro-geometric Approaches to Optimization and Generalization in Mathematical Machine Learning"

8. Algebraic Geometry and Machine Learning Workshop, SIAM Conference on Mathematics of Data Science
Oct 25, 2024, Atlanta, GA, USA
"Geometry of Polynomial Neural Networks"

7. Numerical Analysis Seminar, Institute of Mathematical Research and Dept. of Mathematics, The University of Hong Kong
Apr 30, 2024, Hong Kong
"Geometry of Polynomial Neural Networks"

6. Special Session: Applications of Algebra and Geometry, American Mathematical Society 2024 Spring Central Sectional Meeting
Apr 20, 2024, Milwaukee, WI, USA
"An Algebraic Approach to Supply Network Formation and Fragility"

5. Dept. of Mathematics and Statistics, University of Massachusetts Amherst
Apr 12, 2024, online
"Geometry of Polynomial Neural Networks"

4. Level Set Seminar, Dept. of Mathematics, UCLA
Apr 8, 2024, Los Angeles, CA, USA
"Geometry of Polynomial Neural Networks"

3. Applied Algebra Seminar, Dept. of Mathematics, University of Wisconsin–Madison
Dec 2023, Madison, WI, USA
"Geometry of Polynomial Neural Networks"

2. Deep Learning Theory Seminar, UCLA
Sep 2023, Los Angeles, CA, USA
"Neural Collapse Beyond Unconstrained Feature Models"

1. SCISS, UCLA
Jul 2023, Los Angeles, CA, USA
"Mathematical Machine Learning: Theory & Beyond"

Contributed Talks

6. Workshop on Real Applied Algebra
Mar 23–25, 2026, University of Copenhagen, Denmark
"LLC in transformers: An algebro-geometric lens on generalization in neural networks"

5. Higher-order interactions at the crossroads of geometry and topology
Dec 10–12, 2025, Paris, France
"Geometry of Higher-Order Structures in Machine Learning"

4. Matroids, Rigidity, and Algebraic Statistics, ICERM Workshop
Mar 17, 2025, Providence, RI, USA
"Geometric Framework for Optimizing Algebraic Neural Networks"

3. 2024 UC Davis Peter Hall Conference on Statistics in the Age of AI
Nov 9, 2024, Davis, CA, USA
"Optimization in Polynomial Neural Networks: Insights from Algebraic Geometry"

2. Women in Algebraic Statistics Workshop, St John's College, University of Oxford
Jul 12, 2024, Oxford, UK
"Algebraic Structures in Terminal Phase of Neural Network Training"

1. Institute for Mathematical and Statistical Innovation (IMSI)
Oct 10, 2023, Chicago, IL, USA
"Simplex ETF – Algebraic structure arising in the terminal phase of neural network training"

Teaching

Summary: Instructor, Teaching Associate, and Teaching Assistant for 25+ course offerings across statistics, engineering, and machine learning, reaching 1000+ students at UCLA and the University of Oxford.

Instructor of record: Prep Camp, Master of Applied Statistics & Data Science (MASDS), UCLA, 2021–2023

Teaching Associate / Teaching Assistant, UCLA, 2020–2023:
Statistics and Data Science (STATS 13, ENGR 116)
Engineering Systems (ENGR 200, 202, 203, 205)

Teaching Assistant, Oxford Machine Learning Summer School

Service & Outreach

Association for Computing Machinery (ACM) XRDS Magazine
Editor-in-Chief, 2022–2025
Lead Editor, 2021
Feature Editor, 2020

Outreach
Distinguished Women in Statistics and Data Science Workshop Series, 1st (Women in Academia, May 23, 2023) and 2nd Symposium (Women in Industry, Oct 17, 2023)
UCLA Society of Women in Statistics (Committee, 2022–2024)
UCLA Statistics Club (Mentor, 2022–2024)
Invited speaker: Rotary Club, Westwood Village (2023); UCLA Bruin Professionals (2024); Dublin High School, Dublin, CA (2024)

Reviewing
NeurIPS, ICML, ICLR, Algebraic Statistics, SIAM Journal on Applied Algebra and Geometry, Effective Methods in Algebraic Geometry (MEGA)

Memberships

American Mathematical Society (AMS), Association for Women in Mathematics (AWM), American Statistical Association (ASA), Institute of Mathematical Statistics (IMS), Society for Industrial and Applied Mathematics (SIAM), Association for Computing Machinery (ACM)